

Dashboard Indicators Mapping Document

Rwanda NBSAP Monitoring Dashboard 2025–2030

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Document Control

Field	Value
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Project	Rwanda NBSAP Monitoring Dashboard
Prepared by	NBSAP Dashboard Development Team
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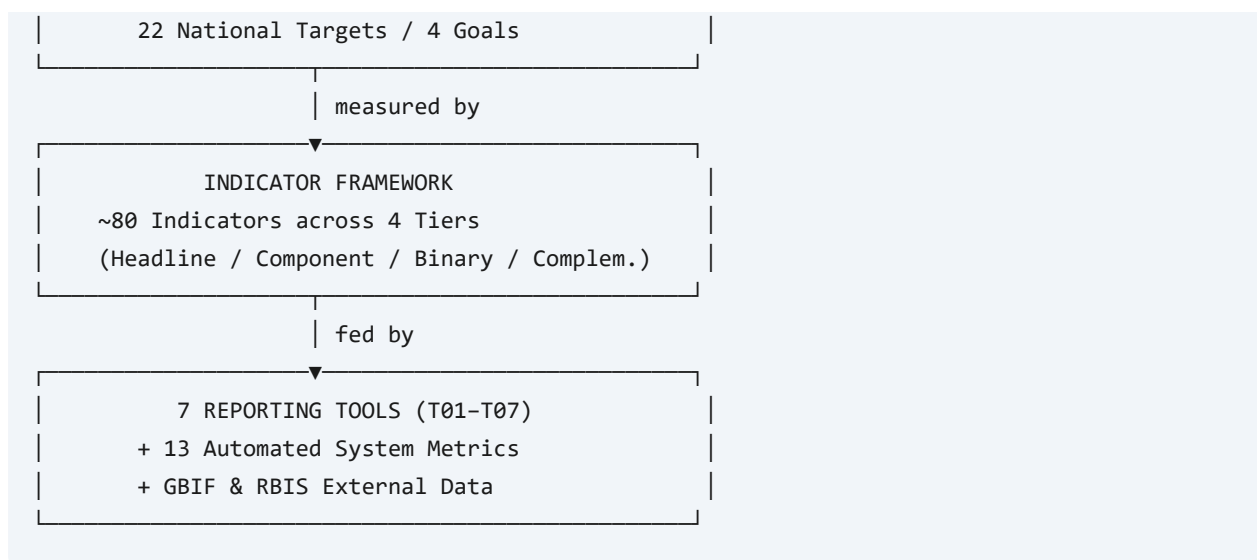
1. Overview

1.1 Purpose

This document maps the complete indicator framework used by the NBSAP Monitoring Dashboard, showing how Rwanda's 22 national targets, ~80 indicators, 7 reporting tools, and 13 automated metrics interconnect. It serves as the definitive reference for understanding what the dashboard measures, how data flows from field reports to national progress, and how the system aligns with the Kunming-Montreal Global Biodiversity Framework (KM-GBF).

1.2 Framework Structure





2. NBSAP Targets by KM-GBF Goal

Goal A — Reduce Threats to Biodiversity (Targets 1–8)

Target	Title	Headline Indicator	Responsible Stakeholders
1	Participatory biodiversity-inclusive spatial land use planning	% of land/water under biodiversity-inclusive spatial plans	REMA, MININFRA, District Authorities, MINAGRI
2	Restoration of degraded ecosystems	Area (ha) of degraded land/water under restoration	REMA, RFA, District Authorities, RNRA
3	Protected area expansion and management	% of national territory under effective protection (PAs + OECMs)	RDB, REMA, District Authorities
4	Species conservation and human-wildlife conflict reduction	% threatened species with improved conservation status	RDB, REMA, Conservation NGOs
5	Sustainable management of wild species	% reduction in illegal harvesting across forests/fisheries/wildlife	MINAGRI, RAB, RFA, REMA
6	Invasive alien species prevention and control	Invasive alien species establishment rate reduction in PAs (%)	REMA, RDB, RFA, MINAGRI
7	Pollution reduction from agriculture, mining and industry	Concentration of key pollutants harmful to biodiversity (index)	REMA, MINAGRI, WASAC, Rwanda Mines Board
8	Climate resilience and nature-based solutions	Area under nature-based solutions addressing climate resilience (ha)	REMA, RFA, District Authorities, MINAGRI

Goal B — Meet People's Needs Through Sustainable Use (Targets 9–12)

Target	Title	Headline Indicator	Responsible Stakeholders
9	Sustainable wildlife benefits for communities	% of PA revenue shared with local communities (TRSP)	RDB, REMA, District Authorities
10	Sustainable agriculture, aquaculture and forestry	% of agricultural/fisheries/forestry operations with biodiversity-inclusive practices	MINAGRI, RAB, RFA, REMA
11	Ecosystem services restoration and maintenance	Value of ecosystem services maintained or restored (USD equivalent)	REMA, RFA, RNRA, District Authorities
12	Urban biodiversity and green space integration	% of urban areas with biodiversity-inclusive green space plans	MININFRA, City of Kigali, REMA, District Authorities

Goal C — Tools and Solutions (Targets 13–17)

Target	Title	Headline Indicator	Responsible Stakeholders
13	Access and benefit sharing from genetic resources	Number of ABS agreements in place (cumulative)	REMA, RDB, MINAGRI
14	Biodiversity mainstreaming across all sectors	% of national sector policies with biodiversity safeguards integrated	REMA, MINECOFIN, All Ministries
15	Business and financial biodiversity risk disclosure	% of large companies/financial institutions reporting biodiversity risks	REMA, BNR, PSF
16	Sustainable consumption and production	National sustainable consumption and production index score	REMA, RBS, MINICOM
17	Biosafety capacity and biotechnology control	Biosafety regulatory framework operationalization score (%)	REMA, RDB, RAB

Goal D — Means of Implementation (Targets 18–22)

Target	Title	Headline Indicator	Responsible Stakeholders
18	Harmful incentive reform and positive biodiversity incentives	Value of harmful subsidies reformed or redirected (RWF millions)	MINECOFIN, REMA, MINICOM
19	Biodiversity finance mobilization (USD 500 million)	Cumulative biodiversity finance mobilised (USD millions)	REMA, MINECOFIN, GCF Focal Point

20	Capacity building, technology transfer and education	Number of biodiversity professionals trained (cumulative)	REMA, RDB, University of Rwanda
21	Biodiversity information systems and data access	Number of biodiversity records accessible via RBIS (thousands)	REMA, RDB, University of Rwanda, NISR
22	Inclusive participation and traditional knowledge	% of biodiversity governance processes with documented inclusive participation	REMA, Civil Society, District Authorities

3. Indicator Tier Classification

The system follows the official KM-GBF three-tier indicator structure:

3.1 Tier Definitions

Tier	Name	Purpose	Example
Tier 1	Headline Indicators	High-level, internationally comparable metrics required for CBD national reporting	% of national territory under protection
Tier 2	Component Indicators	Detailed implementation metrics for adaptive management	Number of hectares reforested per district
Tier 3a	Binary Indicators	Yes/No policy condition assessments	National ABS law enacted (Yes/No)
Tier 3b	Complementary Indicators	Additional context metrics supporting analysis	Species occurrence count from GBIF

3.2 Indicator Distribution

Tier	Count	% of Total	Typical Data Source
Headline	~22	28%	Aggregated from T01–T07 reports
Component	~35	44%	Detailed field data from T02–T04
Binary	~10	13%	Policy assessment via T01
Complementary	~12	15%	GBIF, RBIS, analytical calculations
Total	~79	100%	

4. Reporting Tool → Indicator Mapping

4.1 Tool-to-Target Alignment

Tool	Domain	Primary Targets Fed	Key Indicators Updated
T01	National Institutional Reporting	14, 17, 18	Policy integration %, biosafety score, subsidy reform value
T02	Ecosystem & Habitat Monitoring	1, 2, 8, 11	Spatial plan coverage, restoration hectares, NbS area
T03	Protected Area Management	3, 4, 6	PA coverage %, management effectiveness, illegal cases
T04	Human-Wildlife Conflict	4, 5, 9	HWC incident count, species conservation status, community benefits
T05	Finance & Resource Mobilisation	18, 19	Budget allocated, disbursed, finance mobilised
T06	Private Sector EIA Compliance	7, 15	EIA compliance rate, biodiversity risk disclosure
T07	Community Engagement	9, 20, 22	Community participants, traditional knowledge entries

4.2 Data Flow: Report → Metric → Target Progress

```
Reporter submits T02 report
|
|— form_data includes: forest_hectares = 150, wetland_hectares = 45
|
|— Report enters verification queue (status: 'pending')
|
|— Reviewer approves report (status: 'approved')
|
|— Database trigger fires:
|   |— system_metrics['forest_restored_ha'] += 150
|   |— system_metrics['wetland_restored_ha'] += 45
|   |— system_metrics updated_at = NOW()
|
|— Weighted progress trigger fires:
|   |— Target 2 progress = f(T02_weight × T02_metric_value, ...)
|   |— nbsap_targets[2].progress updated
|
|— WebSocket pushes update to all connected dashboards
```

5. Automated System Metrics

5.1 Metric Inventory

The system automatically computes 13 national-level metrics from approved report data:

#	Metric	Unit	Source Tool	Target(s) Fed
1	Forest Area Restored	Hectares	T02	2, 8
2	Wetland Area Restored	Hectares	T02	2, 11
3	Protected Area Coverage	%	T03	3
4	Average Management Effectiveness	Score (0–100)	T03	3
5	Total Illegal Cases Reported	Count	T03	3, 5
6	Total HWC Incidents	Count	T04	4
7	Total Budget Allocated	RWF	T05	19
8	Total Budget Disbursed	RWF	T05	19
9	Budget Disbursement Rate	%	T05 (calculated)	19
10	EIA Compliance Rate	%	T06	7, 15
11	Private Sector Restoration Commitments	Hectares	T06	7, 15
12	Community Participants	Count	T07	22
13	Traditional Knowledge Entries	Count	T07	22

5.2 Metric Computation Rules

Rule	Description
Summation	Forest ha, wetland ha, illegal cases, HWC incidents, budget, participants, TK entries are summed across all approved reports
Average	Management effectiveness is averaged across all approved T03 reports
Ratio	Disbursement rate = (total disbursed / total allocated) × 100
Coverage	PA coverage = sum of reported coverage from T03 reports
Reversal	When an approved report is deleted, its contribution is subtracted from affected metrics

6. Progress Calculation System

6.1 Tool Weights

Each reporting tool has a configurable weight determining its contribution to target progress. Weights are stored in the `tool_weights` table and can be adjusted by REMA Administrators.

Example Weight Configuration for Target 2 (Ecosystem Restoration):

Tool	Weight	Contribution
T02 (Ecosystem Monitoring)	0.50	Primary data source
T03 (Protected Areas)	0.20	Restoration within PAs
T07 (Community Engagement)	0.15	Community-led restoration
T01 (Institutional)	0.10	Policy support
T05 (Finance)	0.05	Financial support
Total	1.00	

6.2 Progress Formula

Target Progress (%) = SUM(tool_weight[i] × tool_completion[i])
for all tools linked to this target

Where:

tool_completion[i] = (current_metric_value / target_metric_value) × 100
capped at 100%

6.3 Status Colour Coding

Colour	Range	Label	Meaning
Green	>= 70%	On Track	Target likely to be met by 2030
Amber	40% – 69%	At Risk	Intervention may be needed
Red	< 40%	Behind	Urgent attention required

7. External Data Integration

7.1 GBIF Data Mapping

GBIF Data	Dashboard Usage	Target Relevance
Species occurrence count (Rwanda)	Total biodiversity records metric	Target 21 (data access)
Recent occurrences (last 24h/7d)	RBIS Dashboard metrics	Target 21
Species with coordinates	Map overlay (biodiversity hotspots)	Targets 3, 4, 6
Scientific names	Species inventory	Target 4

7.2 RBIS Data Stream Mapping

Data Stream	Targets Supported	Indicators Fed
Protected Areas Coverage	3	PA extent, management effectiveness
Threatened Species Monitoring	4	Species conservation status
Forest Cover Change	2, 8	Forest area, restoration progress
Wetland Extent	2, 11	Wetland area, ecosystem services
Species Distribution	4, 21	Occurrence records, data access
Invasive Species Tracking	6	IAS establishment rate
Ecosystem Restoration	2, 8, 11	Restoration hectares
Sustainable Use Indicators	5, 9, 10	Harvesting metrics, community benefits

8. Dashboard Page → Data Source Mapping

Dashboard Page	Data Sources	Metrics Displayed
Executive Dashboard	system_metrics, toolkit_reports (approved), nbsap_targets	All 13 automated metrics, target progress bars, recent reports
National Targets	nbsap_targets, indicators, nbsap_milestones	22 targets with progress, milestones, linked indicators
Indicators	indicators, rbis_linkages	~80 indicators with status, tier, progress, RBIS linkage
Interactive Map	districts, GeoJSON files, GBIF API	30 districts, protected areas, rivers, lakes, species occurrences
Reporting Toolkit	toolkit_reports (user's own)	Submission forms (T01–T07), submission history
Verification Queue	toolkit_reports (pending)	Pending submissions with form data and evidence
Reports & Analytics	toolkit_reports (all), system_metrics	Aggregated charts, trend analysis, export functions
RBIS Dashboard	rbis_data_streams, GBIF API, indicators, rbis_linkages	8 data streams, RBIS metrics, indicator-stream linkages
Compliance	compliance_records, districts	Compliance scores, district status, EIA tracking
Risk Register	risks	Active risks by severity, category, mitigation status

9. Geographic Data Layers

Layer	Source	Format	Data
District Boundaries	rwanda-districts.geojson	GeoJSON	30 district polygons across 5 provinces
Protected Areas	rwanda-protected-areas.geojson	GeoJSON	National parks, forest reserves, wetlands
River Network	rwanda-rivers.geojson	GeoJSON	Major river systems
Lakes	lakes database table	Database	Inland water bodies with coordinates
Species Occurrences	GBIF API (live)	API response	Point locations of biodiversity observations
District Compliance	districts database table	Database	Compliance score and forest cover per district

10. CBD National Report Alignment

The indicator framework enables direct export for Rwanda's periodic National Reports to the CBD Secretariat:

CBD Reporting Requirement	System Coverage	Export Format
Progress toward national targets	22 targets with quantified progress	CSV, PDF, JSON
Headline indicator values	~22 headline indicators with baselines and current values	CSV
Status of biological diversity	Species data via GBIF, habitat data via T02/T03	PDF report
Threats and drivers	Pollution (T06), IAS (T03), HWC (T04), land use (T02)	CSV, PDF
Implementation measures	Institutional actions (T01), finance (T05), capacity (T07)	CSV, PDF
Effectiveness of measures	Management effectiveness (T03), compliance rates (T06)	CSV

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